

The above is a systematic procedure to compute the elements of L and U. We can easily generalize the scheme.

Let us illustrate the procedure by means of the following example.

Example 6 (a) Solve the following system of equations,

$$4x_1 + 5x_2 - 2x_3 = 5.6$$

$$2x_1 + 3x_2 + x_3 = 1.3$$

$$2x_1 + 4x_2 + 4x_3 = 9.5$$

using the decomposition method.

Linear Systems of Equations

287

$$\begin{bmatrix} 4 & 5 & -2 \\ 2 & 3 & 1 \\ 2 & 4 & 4 \end{bmatrix}$$

to search



C -4.33%

(b) Invert $A = \begin{bmatrix} 4 & 5 & -2 \\ 2 & 3 & 1 \\ 2 & 4 & 4 \end{bmatrix}$

using the decomposition method.

- c) Using A^{-1} , solve the system of equations given in (a) above. Compare the two results.

Solution (a) Solution of Equations using L and U.

We know that $LU = A$.

$$\text{So, } \begin{vmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{vmatrix} \begin{vmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{vmatrix} = \begin{bmatrix} 4 & 5 & -2 \\ 2 & 3 & 1 \\ 2 & 4 & 4 \end{bmatrix}$$

Multiplying, we get:

$$\begin{vmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{vmatrix} = \begin{bmatrix} 4 & 5 & -2 \\ 2 & 3 & 1 \\ 2 & 4 & 4 \end{bmatrix}$$

$$\begin{vmatrix} l_{31} u_{11} & l_{31} u_{12} + l_{32} u_{22} & l_{31} u_{13} + l_{32} u_{23} + u_{33} \end{vmatrix} \quad \begin{bmatrix} 2 & 4 & 4 \end{bmatrix}$$

Computing coefficients, we get,

$u_{11} = 4$	$u_{12} = 5$	$u_{13} = -2$
$l_{21} u_{11} = 2$	$l_{21} u_{12} + u_{22} = 3$	$l_{21} u_{13} + u_{23} = 1$
or, $l_{21} = \frac{2}{u_{11}} = \frac{2}{4} = \frac{1}{2}$	$u_{22} = \frac{1}{2}$	$u_{23} = 2$
$l_{31} u_{11} = 2$	$l_{31} u_{12} + l_{32} u_{22} = 4$	$l_{31} u_{13} + l_{32} u_{23} + u_{33} = 4$
or, $l_{31} = \frac{2}{u_{11}} = \frac{2}{4} = \frac{1}{2}$	$l_{32} = 3$	$u_{33} = -1$

It follows that $A = LU$

$$= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 3 & 1 \end{bmatrix} \times \begin{bmatrix} 4 & 5 & -2 \\ 0 & \frac{1}{2} & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

Since $Ly = b$, we have

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5.6 \\ 1.3 \\ 9.5 \end{bmatrix}$$

Multiplying the matrices on the left-hand side, we get,

$$y_1 = 5.6$$

$$\frac{1}{2} y_1 + y_2 = 1.3$$

$$\frac{1}{2} y_1 + 3y_2 + y_3 = 9.5$$

Solving the above system, we get,

$$y_1 = 5.6, y_2 = -1.6 \text{ and } y_3 = 11.2$$

Using $Ux = y$, we get,

$$\begin{bmatrix} 4 & 0 & -2 \\ 0 & \frac{1}{2} & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5.6 \\ -1.5 \end{bmatrix}$$

Using $Ux = y$, we get,

$$\begin{bmatrix} 4 & 0 & -2 \\ 0 & \frac{1}{2} & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.6 \\ -1.5 \\ 11.2 \end{bmatrix}$$

Multiplying the matrices on the left-hand side, we get,

$$\begin{aligned} 4x_1 + 5x_2 - 2x_3 &= 5.6 \\ \frac{1}{2}x_2 + 2x_3 &= -1.5 \\ -x_3 &= 11.2 \end{aligned}$$

Solving the above system of equations by backward substitution, we get,

$$x_3 = -11.2, x_2 = 41.8 \text{ and } x_1 = -56.5$$

(b) The matrices L and U have already been computed in (a) above.

Now, $LL^{-1} = I$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) The matrices L and U have already been computed in (a) above.

Now, $LL^{-1} = I$

$$\text{Let } L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ l'_{21} & 1 & 0 \\ l'_{31} & l'_{32} & 1 \end{bmatrix}$$

Linear Systems of Equations

289

$$\text{Therefore, } \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ l'_{21} & 1 & 0 \\ l'_{31} & l'_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiplying both matrices and comparing like terms, we get:

Multiplying both matrices and comparing like terms, we get:

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} + l'_{21} & 1 & 0 \\ \frac{1}{2} + 3l'_{21} + l'_{31} & 3 + l'_{32} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{1}{2} + l'_{21} = 0; \quad l'_{21} = -\frac{1}{2}$$

$$\frac{1}{2} + 3l'_{21} + l'_{31} = 0; \quad l'_{31} = 1$$

$$3 + l'_{32} = 0; \quad l'_{32} = -3$$

Substituting values in L^{-1} , we get,

$$L^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 1 & -3 & 1 \end{bmatrix}$$

Also, $UA^{-1} = L^{-1}$

$$\begin{bmatrix} 4 & 5 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \end{bmatrix}$$

Also, $UA^{-1} = L^{-1}$

$$\begin{bmatrix} 4 & 5 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 1 & -3 & 1 \end{bmatrix}$$

Multiplying and comparing coefficients, we get

$$a'_{31} = -1; \quad a'_{32} = 3; \quad a'_{33} = -1$$

$$a'_{21} = 3; \quad a'_{22} = -10; \quad a'_{23} = 4$$

$$a'_{11} = -4; \quad a'_{12} = 14; \quad a'_{13} = -\frac{11}{2}$$

Substituting values in A^{-1} , we get,

$$A^{-1} = \begin{bmatrix} -4 & 14 & -\frac{11}{2} \\ 3 & -10 & 4 \\ -1 & -3 & -1 \end{bmatrix}$$

290

Numerical Analysis with C++

The system of equations can be solved using: $x = A^{-1}b$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 & 14 & -\frac{11}{2} \\ 3 & -10 & 4 \\ -1 & -3 & -1 \end{bmatrix} \begin{bmatrix} 5.6 \\ 1.3 \\ 11.2 \end{bmatrix} = \begin{bmatrix} -56.5 \\ 41.8 \\ -11.2 \end{bmatrix}$$

The solution is as follows:

$$x_1 = -56.5, x_2 = 41.8 \text{ and } x_3 = -11.2$$

Both results are the same.

8.6 TRIANGULAR DECOMPOSITION FOR SYMMETRIC MATRICES

When the given matrix A is real symmetric and positive definite, then we may decompose A as:

$$A = LL^T$$

$$A^{-1} = (LL^T)^{-1} = (L^T)^{-1} (L^{-1})$$